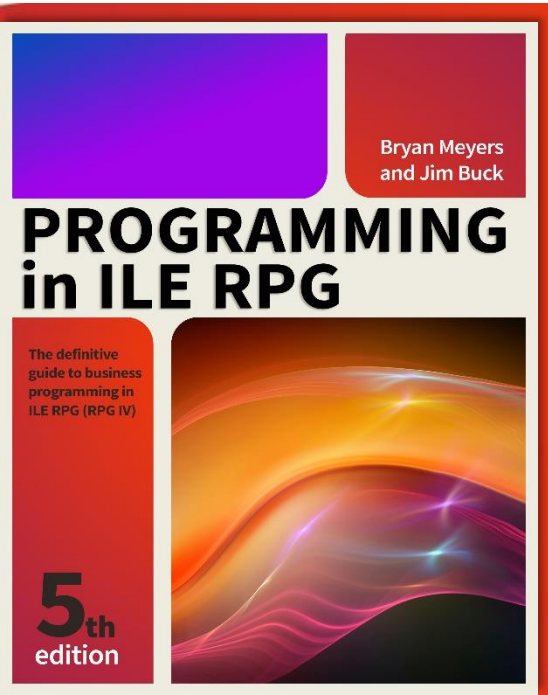
Chapter 4:
Using Declarations



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Introducing Declarations

- Program must declare all files and indicate how it will use them
  - When a file is externally defined, all of its records and fields are automatically available to the program and do not have to be redefined in the RPG program itself
- Program must define all other data items and specify data properties
  - Such as items to be calculated, indicators, constants
- Declarations follow control options
- Six declaration instructions
  - Dcl-f (Declare file)
  - Dcl-c (Declare constant)
  - Dcl-s (Declare standalone variable)
  - Dcl-ds (Declare data structure)
  - Dcl-pr (Declare prototype) -- covered in Advanced RPG
  - Dcl-pi (Declare procedure interface) – covered in Advanced RPG

Declaring Files

- File name
- Device
  - **Disk – database file (default)**
  - **Printer – printer file**
  - Workstn – display file – covered later in this course

```
// Externally described files
Dcl-f Customers Disk;
Dcl-f Custlist  Printer Oflind(Endofpage);
```

```
// Program described files- more on these at end of course with “legacy” fixed form
Dcl-f Customers Disk(145);
Dcl-f Qprint    Printer(132) Oflind(*Inof);
```

File Usage

- Usage(*Input)
  - Program can read existing records from file
  - Default for Disk device
- Usage(*Output)
  - Program can write new records to file
  - Default for Printer device
- Usage(*Update) – covered in Advanced RPG
  - Program can read, change existing records
  - But not delete
  - Implies Usage(*Input) as well
- Usage(*Delete) – covered in Advanced RPG
  - Program can delete existing records

File Usage

- Usage(*Input:*Output) – later in this course
  - Program can read records and add new ones
  - Default for Workstn device – covered later in this course
- Usage(*Update:*Delete) – Advanced RPG
  - Program can read, change or delete existing records
- Usage(*Update:*Delete:*Output) – Advanced RPG
  - Program can read, change, delete, or add new records

```
Dcl-f Customers Disk    Usage(*Update:*Delete:*Output);
Dcl-f Custlist  Printer Usage(*Output) Oflind(Endofpage);
```

File Keywords

- Keyed
  - Processes database records in keyed order
  - For program described file, must also specify key length and possibly location
- Oflind
  - Names page **overflow indicator** for printer file

```
// Externally described files
Dcl-f Customers Disk    Keyed;
Dcl-f Custlist  Printer Oflind(Endofpage);
```

```
// Program described files (more later)
Dcl-f Customers Disk(145)    Keyed(*Char:5) Keyloc(2);
Dcl-f Qprint    Printer(132) Oflind(*Inof);
```

File Keywords
(valid information, not directly practiced in this course)

- Rename
  - Renames record format
- Prefix
  - Renames all fields in record format

```
Dcl-f Customers;
Dcl-f Custnames Rename(Custrec:Custrec2) Prefix(C2);
```

File Keywords
(valid information, not directly practiced in this course)

- Extdesc
  - Names file that compiler uses to retrieve the record layouts
    - If different from file name, or to specify library
  - Overrides file name at compile time
  - Valid only for externally described files
- Extfile, Extmbr

```
Dcl-f Customers Extdesc('DEVLIB/CUSTOMERS')
                Extfile('PRDLIB/CUSTOMERS') Extmbr('DOMESTIC');
```

```
Dcl-f Customers Extdesc('DEVLIB/CUSTOMERS') Extfile(*EXTDESC);
```

Numeric Literals



- Contain only digits 0-9, radix (decimal) character, and leftmost sign
- No blanks, currency symbols, percent signs, or thousands separators (commas)
- **Not** quoted
- Can be up to 63 digits, 63 decimal positions

```
-401230.12
0.0715
102
1
+3
-1
3.1416
.123456789
```

Character literals

- Character strings
- Can contain any character, including blanks and nondisplay characters
- **Must be quoted**
  - Enclosed between apostrophes (')
- Can be up to 16,380 bytes

```
'John Doe'
'Abc 246 #18w'
'321444'
'45%'
```

Typed literals

- Used to express data values other than character or numeric
- Quoted value, preceded by data type code



Data Type	*Typed Literal*
Date	`D'2008-03-15'`
Time	`T'08.56.20'`
Timestamp	`Z'2008-03-15-08.56.20.000000'`
Hexadecimal	`X'F0F0F0'`

Declaring Named Constants

- Dcl-**c** instruction declares named **constant**
  - Associates data item name with literal value
- Can use named constant whenever literal is appropriate
  - Value cannot change



```
Dcl-c Taxrate     .0765;
Dcl-c Pi          3.142;
Dcl-c Country     'United States';
Dcl-c Longestword 'floccinaucinihilipilification';
Dcl-c Disclaimer  'This report is proprietary, +
                   and may contain confidential information, +
                   not to be disclosed.';
Dcl-c Phoneedit   '0(   )   -    ';
Dcl-c Millennium  D'2001-01-01';
```

Figurative Constants

- *Blank (or *Blanks)
- *Zero (or *Zeros)
- *Off
- *On
- *Hival
- *Loval
- *All
- *Null



Defining Standalone Variables

- Dcl-s instruction declares standalone variable
- Data item name
- Data type, and if applicable, size

```
Dcl-s Citystatezip          Char(40);
Dcl-s Taxexempt             Ind;
Dcl-s Totaldue              Packed(7:2);
Dcl-s Yeartodatetotalsales  Packed(11:2);
Dcl-s Counter               Uns(10);
Dcl-s Basedate              Date;
```



Character Data

- Char
  - Fixed length character string
  - Length can be up to 16,773,104 bytes
- Varchar
  - Variable length character string
    - Used when length is uncertain
    - Provide maximum size in declaration, such as 256 in this example
  - Maximum size up to 16,773,100 bytes

```
Dcl-s Address Char(35);
Dcl-s Email   Varchar(256);
```

Character Data - Indicator

- Ind (Indicator)
  - Signals true/false state --
  - Program sets indicator *On ('1') or *Off ('0')
  - Single byte
    - Length not specified

```
Dcl-s Endofpage Ind;
```

Numeric Data

- Maximum precision is 63 digits
  - Maximum scale is 63 decimal places behind the decimal
- Zoned
  - Each digit uses one byte
  - Sign is embedded in rightmost byte
- Packed view this: http://screencast.com/t/kYzgeK2HxDz
  - Each digit uses ½ byte (“nybble”)
  - Sign uses additional ½ byte
  - Most efficiently used with odd number of digits

```
Dcl-s Daynumber Zoned(2:0);
Dcl-s Totaldue  Packed(7:2);
```

Numeric Data

- Integers are most compact numeric format
  - Defined as 3, 5, 10, or 20 digits
  - Occupy 1, 2, 4, or 8 bytes
  - No decimal places
- Int
  - Signed integer
- Uns
  - Unsigned integer



```
Dcl-s Range   Int(5);
Dcl-s Counter Uns(5);
```

Numeric Data

- Integer range depends upon size

Data Type	*LOVAL	*HIVAL
Int(3)	-128	127
Int(5)	-32,768	32,767
Int(10)	-2,147,483,648	2,147,483,647
Int(20)	-9,223,372,036,854,775,808	9,223,372,036,854,775,807
Uns(3)	0	255
Uns(5)	0	65,535
Uns(10)	0	4,294,967,295
Uns(20)	0	18,446,744,073,709,551,615

Date and Time Data

- Date
- Time
- Timestamp
  - May include fractional seconds
  - Defaults to microseconds



```
Dcl-s Today      Date;
Dcl-s Now        Time;
Dcl-s Lastupdate Timestamp(0);  // 2015-06-30-12.34.56
Dcl-s Current    Timestamp;     // 2015-06-30-12.34.56.789012
```

Assigning Initial Value

- Inz (Initialize) keyword specifies initial value for data item
  - Value of the standalone variable could be changed later by logic in the program
- If no Inz keyword is present
  - Character data defaults to blanks
  - Numeric data defaults to zeros
  - Date data defaults to *LOVAL

```
Dcl-s Maxlimit Packed(7:0) Inz(10500);
Dcl-s Compname Char(40)    Inz('Kay Elmnop Enterprises');
Dcl-s Check    Char(9)     Inz(X'0000000000000000'); // hexadecimal initializer
Dcl-s Basedate Date        Inz(D'1899-12-30');
```

Assigning Initial Value

- Inz may use reserved words or other constants
  - Inz(*Blanks)
  - Inz(*Zeros)
  - Inz(*Loval)
  - Inz(*Hival)
  - Inz(*Sys) for system date
  - Inz(*Job) for job date
  - Inz(*User) for user ID

```
Dcl-s Highlimit Packed(7:2) Inz(*Hival);
Dcl-s Today     Date        Inz(*Sys);
Dcl-s Username  Char(10)    Inz(*User);
```

Defining Data Structures

- Data structure groups variables into logical structure
- RPG can process entire data structure or individual subfields
- Dcl-ds instruction declares data structure
  - End-ds instruction ends declaration
- Dcl-subf instruction declares subfields
  - Instruction is usually optional and rarely used



```
Dcl-ds Phone;
  Areacode Zoned(3:0);
  Exchange Zoned(3:0);
  Localnbr Zoned(4:0);
  Name     Char(35);
End-ds;
```

Defining Data Structures

- Data structure is considered to be character variable
  - Len (Length) keyword may specify data structure length
  - If you leave out the Len keyword, the program automatically figures out the length, therefore Len is not commonly used
- Subfields are normally contiguous
  - Pos (Position) keyword may specify subfield beginning location

```
Dcl-ds Indicators Len(99);
  Exit   Ind Pos(3);
  Cancel Ind Pos(12);
  Error  Ind Pos(50);
  Sflclr Ind Pos(90);
  Sfldsp Ind Pos(91);
  Sflend Ind Pos(92);
End-ds;
```

Initializing Data Structures

- Inz keyword initializes entire data structure or data structure subfields
  - Inz used with Dcl-ds initializes all subfields with values appropriate to data type
    - Otherwise data structure is initialized to blanks
  - May also initialize subfields individually

```
Dcl-ds Phone Inz;
  Areacode Zoned(3:0);
  Exchange Zoned(3:0) Inz(555);
  Localnbr Zoned(4:0);
  Name     Char(35);
End-ds;
```

Overlapping Subfields

- Overlay keyword allows subfields to overlap

```
Dcl-ds *N Inz;     // *N indicates unnamed data structure
  Phone      Zoned(10:0);
    Areacode Zoned(3:0) Overlay(Phone);
    Exchange Zoned(3:0) Overlay(Phone:4);
    Localnbr Zoned(4:0) Overlay(Phone:7);
End-ds;
```

```
Dcl-ds *N Inz;
  Phone      Zoned(10:0);
    Areacode Zoned(3:0) Overlay(Phone);
    Exchange Zoned(3:0) Overlay(Phone:*Next);
    Localnbr Zoned(4:0) Overlay(Phone:*Next);
End-ds;
```



```
Dcl-ds *N;     // *N indicates unnamed data structure
  FirstName Char(15);
    FirstInitial  Char(1) Overlay(FirstName);
End-ds;
```

Overlapping Subfields

- Overlay may redefine and/or rename subfields

```
Dcl-ds *N Inz;
  Basedate Date;
  Basechar Char(10) Overlay(Basedate);
End-ds;
```



Overlapping Subfields

- Pos keyword allows subfields to overlap relative to entire data structure
- This is a newer keyword that became available with the “totally free” declaration statements

```
Dcl-ds *N Inz;
  Phone      Zoned(10:0);
    Areacode Zoned(3:0) Pos(1);
    Exchange Zoned(3:0) Pos(4);
    Localnbr Zoned(4:0) Pos(7);
End-ds;
```

Externally Described Data Structures (valid information, not directly practiced in this course)

- Ext (External) and Extname (External Name) keywords declare data structure based upon database file layout
  - No need to explicitly declare subfields
- Extfld (External Field) keyword may rename database fields
- May add subfields to externally described data structure

```
Dcl-ds Customers Ext End-ds;
```

```
Dcl-ds Custds Extname('CUSTOMERS');
  Address  Extfld('CADDR');
  Lastdate Date;
  Rating   Char(1);
End-ds;
```

Qualified Data Structures

- Qualified data structure specifies program will use qualified subfield names
  - Identifies source of subfield
  - Data structure must be named
  - May be program described or externally described

```
Dcl-ds Vendor Qualified;
  Company   Zoned(2:0);   // Vendor.Company is how this subfield would be referenced
  Id        Char(7);        // Vendor.Id
  Name      Char(35);       // Vendor.Name
  Address   Char(35);       // Vendor.Address
  City      Char(21);       // Vendor.City
  State     Char(2);        // Vendor.State
  Postcode  Char(10);       // Vendor.Postcode
End-ds;
```

Qualified Data Structures

- Allows identically named subfields in different data structures
  - Qualified subfield name must be unique

```
Dcl-ds Customer Qualified;
  Company   Zoned(2:0);
  Id        Zoned(5:0);
  Name      Char(35);
  Credlimit Zoned(11:2);
End-ds;

Dcl-ds Vendor Qualified;
  Company   Zoned(2:0);
  Id        Char(7);
  Name      Char(35);
End-ds;
```

Using LIKE, LIKEDS, and LIKEREC

- Keywords define new data item with same properties as parent data item
  - Like keyword defines data item with same length and data type as parent
  - Likeds (Like Data Structure) keyword defines data structure or subfield with same subfields as another data structure
  - Likerec (Like Record Format) keyword defines data structure or subfield with same layout as record format in declared file

Using LIKE

- Declared data item adopts length and data type of parent
- May adjust size of new item

```
Dcl-s Statetax  Zoned(7:7);
Dcl-s Fedtax    Like(Statetax);          // Statetax is Zoned(7:7)

Dcl-s FirstName Char(20);
Dcl-s Longname  Like(Firstname:+15);     // Longname is Char(35)

Dcl-ds Customer;
  Name     Like(Longname);               // Name is Char(35)
  Address  Char(35);
  City     Char(21);
  *N       Char(1);                      // *N indicates unnamed subfield
  State    Char(2);
End-ds;
```

Using LIKEDS

- Subfields in new data item are identical to parent data structure
  - Additional subfields not allowed
- New data structure is implicitly qualified
  - Qualified keyword not allowed
- End-ds not used by new data structure

```
Dcl-ds Customer Qualified;
  Name        Char(35);
  Address     Char(35);
  City        Char(21);
End-ds;

Dcl-ds Vendor Likeds(Customer);
```

Using LIKEDS

- Inz(*Likeds) ensures new data structure is initialized same as parent

```
Dcl-ds Customer Qualified;
  …
  Origdate   Date Inz(*Sys);
End-ds;

Dcl-ds Vendor Likeds(Customer) Inz(*Likeds);
```

Using LIKEREC

- Subfields in data item are identical to fields in record layout
  - Must refer to record format in Dcl-f instruction
  - May restrict data item to key fields from record layout
- New data structure is implicitly qualified
  - Qualified keyword not allowed
- End-ds not used

```
Dcl-f Customers Keyed;

Dcl-ds Before   Likerec(Custsrec);
Dcl-ds After    Likerec(Custsrec);

Dcl-ds Custkeys Likerec(Custsrec:*Key)
```

Legacy Code: File Specifications

Glance through this section- covered later in course

- File name
- Usage
  - I = Usage(*Input)
  - O = Usage(*Output)
  - U = Usage(*Update:*Delete)
  - C = Usage(*Input:*Output)
- File designation (input files only)

```
FFilename++IPEASFRlen+LKlen+AIDevice+.Keywords+++++++++++++++++++++++++++++++
FCustomers UF A E             Disk
FCustlist  O    E             Printer Oflind(Endofpage)
```

Legacy Code: File Specifications

- File addition
  - A = Allow program to write new records
- File format
  - E = Externally described
- Device
  - Disk
  - Printer
- Keywords

```
FFilename++IPEASFRlen+LKlen+AIDevice+.Keywords++++++++++++++++++++++++++++++
FCustomers UF A E             Disk
FCustlist  O    E             Printer Oflind(Endofpage)
```

Legacy Code: Definition Specifications

- Data item name
- Definition type
  - C = Constant
  - S = Standalone variable
  - DS = Data structure header
  - (blank) = Data structure subfield

```
DName++++++++++++ETDsFrom+++To/Len+IDc.Keywords+++++++++++++++++++++++++++++++

D Taxrate         C                   .0765
D Country         C                   'United States'

D Totaldue        S             7P 2
```

Legacy Code: Definition Specifications

- Data item name
- Definition type
  - C = Constant
  - S = Standalone variable
  - DS = Data structure header
  - (blank) = Data structure subfield

```
DName++++++++++++ETDsFrom+++To/Len+IDc.Keywords+++++++++++++++++++++++++++++++

D Customer        DS
D   Company                       2S 0
D   Name                         35A
D   City                         21A
D   State                         2A
```

Legacy Code: Definition Specifications

- Data item length
- Data type
  - A = Char
  - N = Ind
  - S = Zoned
  - P = Packed
  - I = Int
  - U = Uns
  - D = Date
  - T = Time
  - Z= Timestamp
- Decimal scale
- Keywords

```
DName++++++++++++ETDsFrom+++To/Len+IDc.Keywords+++++++++++++++++++++++++++++++
```

Legacy Code: Definition Specifications

```
DName++++++++++ETDsFrom+++To/Len+IDc.Keywords+++++++++++++++++++++++++++++++

D Taxrate         C                   .0765
D Country         C                   'United States'

D Totaldue        S                 7P 2

D Customer        DS
D   Company                          2S 0
D   Id                               5S 0
D   Name                            35A
D   Address                         35A
D   City                            21A
D   State                            2A
D   Postcode                        10A
D   Credlimit                       11S 2 Inz(500)
```